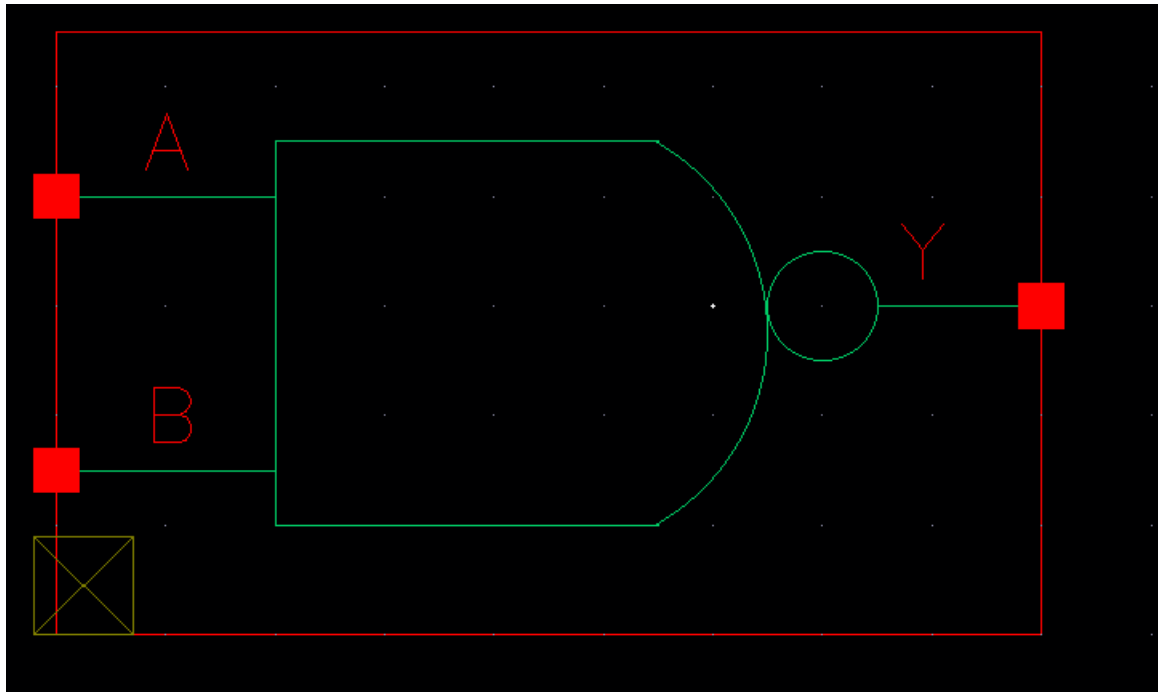


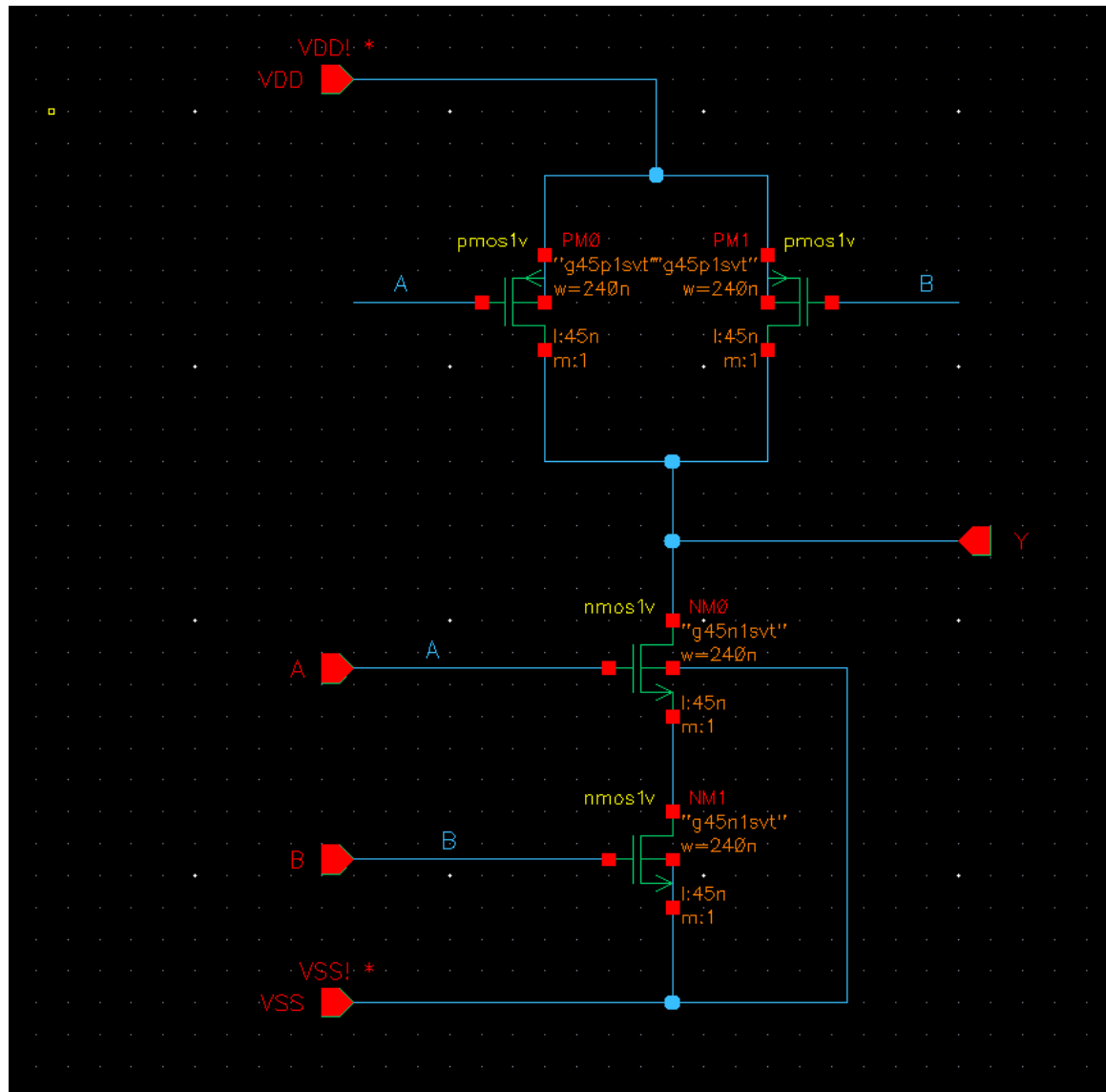
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Library Name:	cgm	
Cell Name:	NAND2XI	
Function/Truth Table:		
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0
Propagation Delay (path based): with extracted parasitic		
Transition		Delay (s)
A → Y ↑ (T_PLH)		1.122n
A → Y ↓ (T_PHL)		1.687n
B → Y ↑ (T_PLH)		1.214n
B → Y ↓ (T_PHL)		1.692n
Output Rise Time (path based):		
Transition		Rise Time (s)
A → Y ↑		2.438n
B → Y ↑		2.436n
Output Fall Time (path based):		
Transition		Fall Time (s)
A → Y ↓		3.243n
B → Y ↓		3.244n

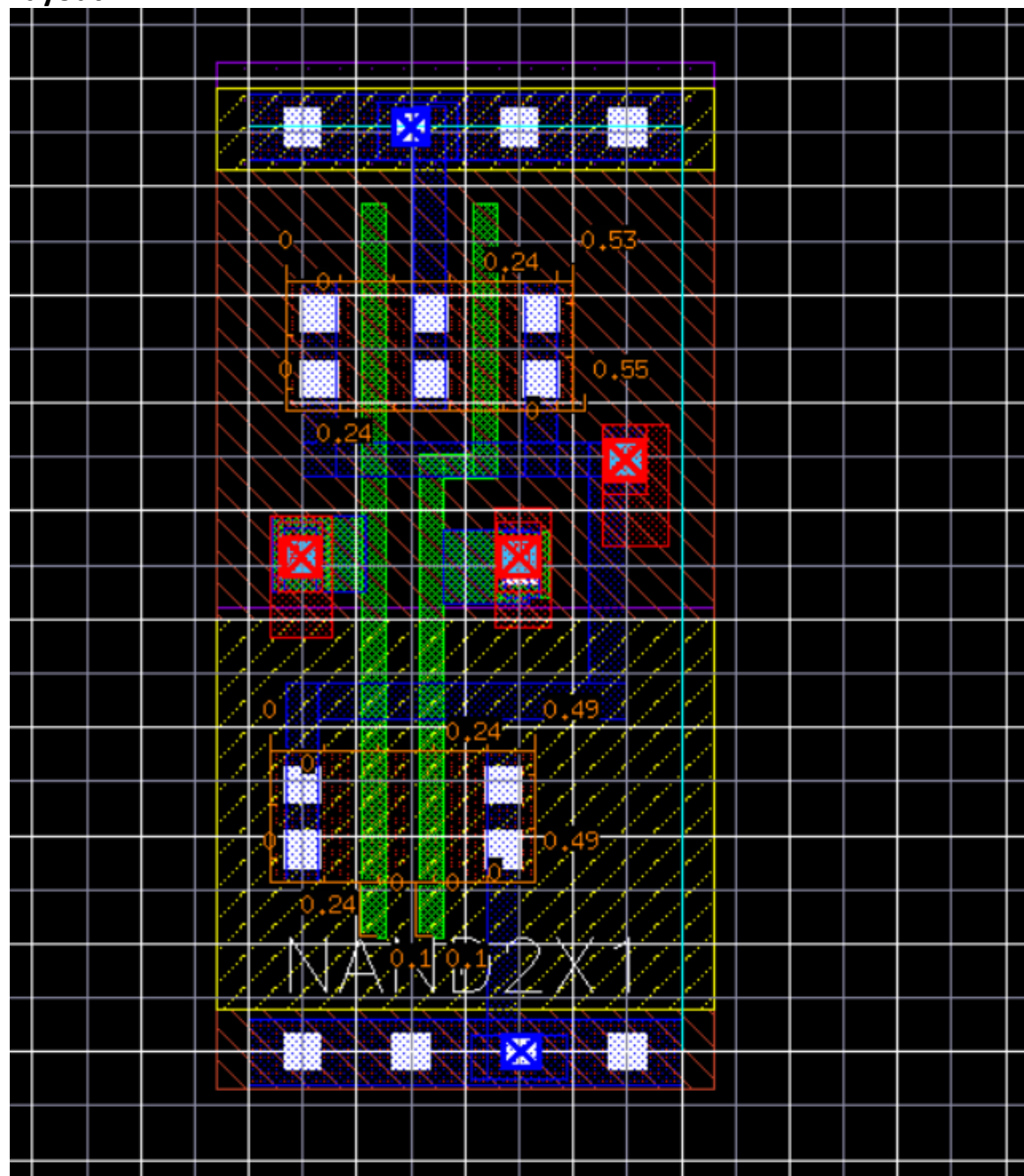
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Layout Height: 1.71 μm **Layout Width: 0.8 μm** **Layout Area: 1.368 μm^2** **Symbol with Port Names:**

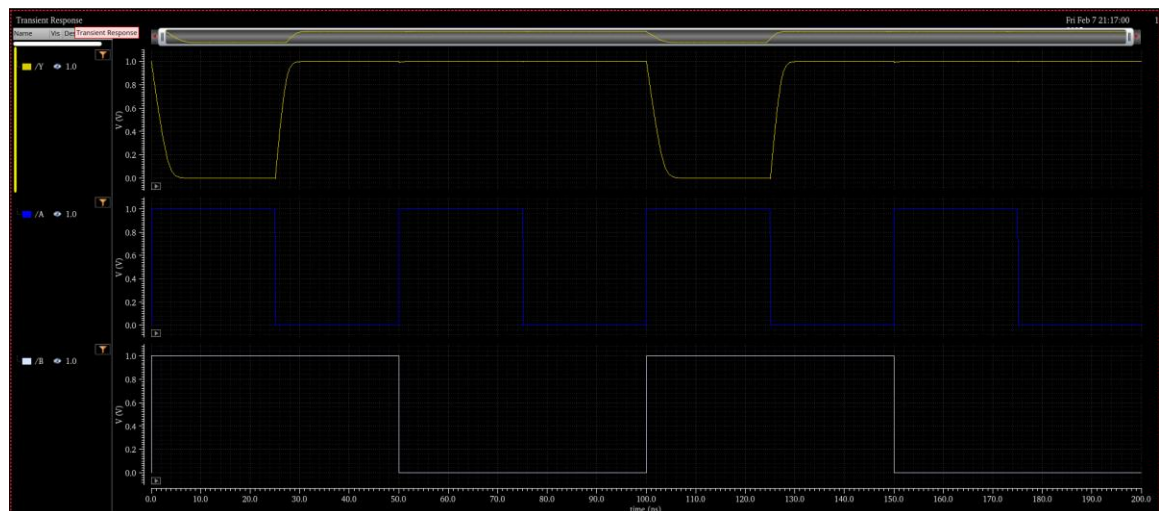
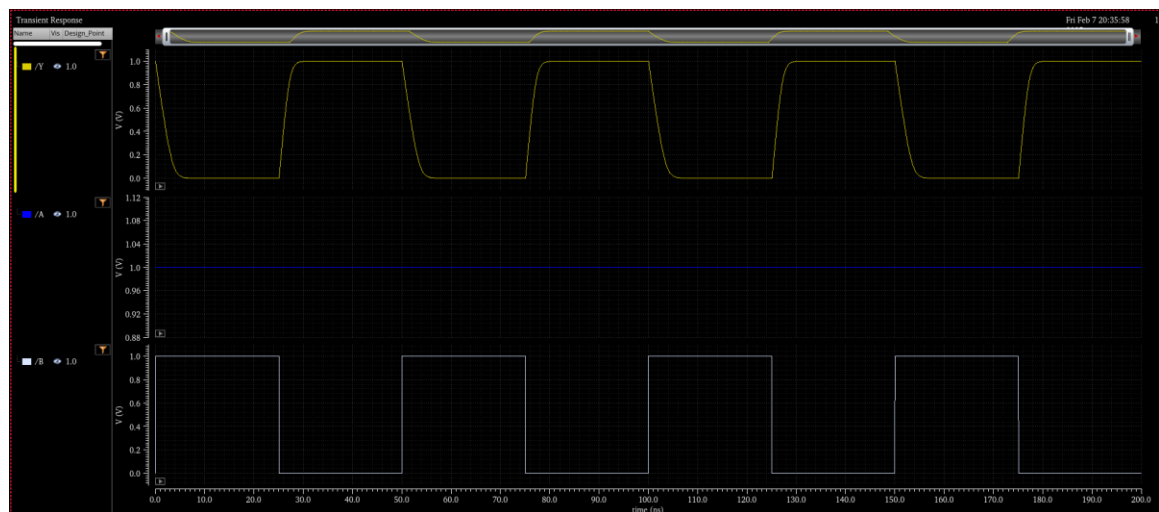
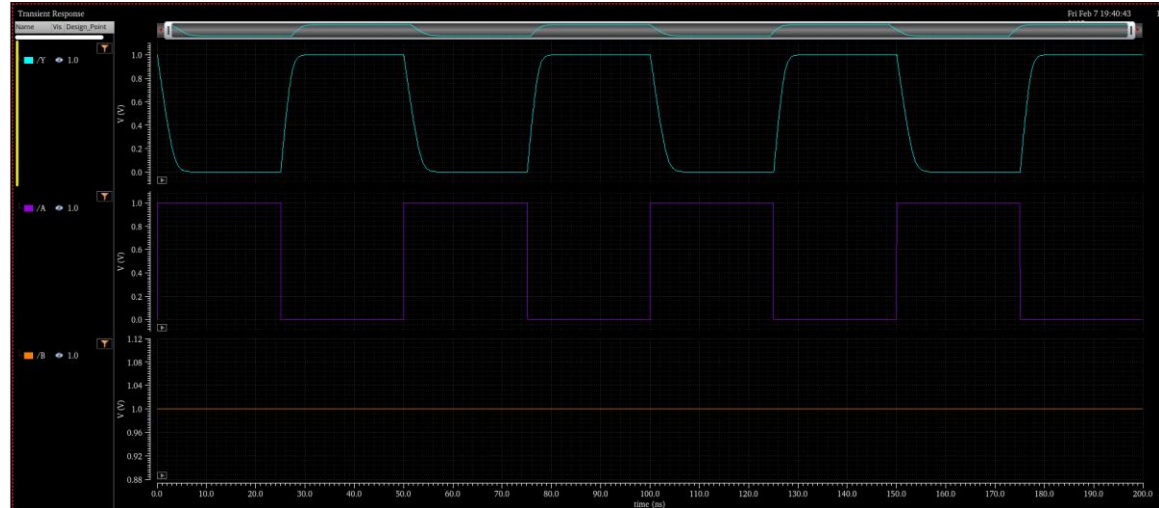
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Schematic:

Layout:



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Functional Simulation Waveforms

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Comments/Notes:

The NAND2X1 gate with extracted parasitics was successfully tested, confirming correct functional verification. However, the timing results show an increase in delays compared to the pre-layout simulation, indicating the impact of parasitic resistances and capacitances from interconnects and device structures.